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Course: Calculus Beta

Assignment: MyLab Math - Uge 5

Does the sequence $\{a_n\}$ converge or diverge? Find the limit if the sequence is convergent.

$$a_n = \sin \left(2\pi - \frac{1}{n} \right)$$

The continuous function theorem for sequences states that if $a_n \rightarrow L$, where $\{a_n\}$ is a sequence of real numbers, and if f is a function that is continuous at L and defined at all a_n , then $f(a_n) \rightarrow f(L)$.

Therefore, we can first consider the behavior of the sequence $\{b_n\}$ where $b_n = \left(2\pi - \frac{1}{n} \right)$. Find the limit of this sequence as n approaches infinity.

$$\lim_{n \rightarrow \infty} \left(2\pi - \frac{1}{n} \right) = 2\pi$$

Now apply the continuous function theorem to find the limit of the original sequence $a_n = \sin \left(2\pi - \frac{1}{n} \right)$.

$$\lim_{n \rightarrow \infty} \sin \left(2\pi - \frac{1}{n} \right) = 0$$

Therefore, the sequence is convergent and converges to 0.